# Answers: Arithmetic on algebraic fractions

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**Summary**

Answers to questions relating to the guide on arithmetic on algebraic fractions.

*These are the answers to Questions: Arithmetic on algebraic fractions.*

**Please attempt the questions before reading these answers!**

## Q1

1.1. $\frac{7}{x}$

1.2. $\frac{2x-1}{3}$

1.3. $\frac{3}{y}$

1.4. $\frac{1}{x}$

1.5. $-1$

1.6. $\frac{3}{2x}$

1.7. $\frac{7}{15y}$

1.8. $\frac{x}{2}$

1.9. $\frac{5x+2}{x(x+1)}$

1.10. $\frac{6}{t-1}$

1.11. $\frac{6}{x-3}$

1.12. $\frac{x-6}{(x+2)(x-2)}$

1.13. $\frac{3c+1}{c^2}$

1.14. $\dfrac{x(2x-3)}{(x-3)(x+3)}$

1.15. $\dfrac{x^2-x-5}{(x+2)^2}$

## Q2

2.1. $\dfrac{2x}{15}$

2.2. $6$

2.3. $\dfrac{x(x+1)}{6}$

2.4. $\dfrac{3}{4}$

2.5. $\dfrac{x-3}{4}$

2.6. $-\dfrac{6}{x}$

2.7. $\dfrac{m}{2}$

2.8. $\dfrac{x+2}{x+3}$

2.9. $\dfrac{3}{4}$

2.10. $\dfrac{x+3}{x+1}$

2.11. $-\dfrac{3}{2}$

2.12. $\dfrac{2x}{3(x-2)}$

## Q3

3.1. $\dfrac{5}{2}$

3.2. $6$

3.3. $\dfrac{1}{2}$

3.4. $\dfrac{3x}{2}$

3.5. $\dfrac{4(x-2)}{3x}$

3.6. $\dfrac{v-1}{v}$

3.7. $-6$

3.8. $2$

3.9. $\frac{1}{2}$

3.10. $\frac{x+3}{x-3}$

3.11. $2$

3.12. $\frac{3(x+2)}{2x}$

## Q4

4.1. $\frac{x+y}{y}$

4.2. $-\frac{6x}{x+2}$

4.3. $\frac{1}{2}$

4.4. $\frac{t+2}{3t}$

4.5. $-\frac{2x(x+1)}{x-1}$

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## Version history and licensing

v1.0: initial version created 12/25 by Donald Campbell as part of a University of St Andrews VIP project.

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